

value should be selected to ensure that the wing is large enough for all flight conditions. Don't forget to convert all wing loadings to takeoff conditions prior to comparisons.

A low wing-loading will always increase aircraft weight and cost. If a very low wing-loading is driven by only one of the requirements, a change in design assumptions (such as a better high-lift system) may allow a higher wing-loading.

Also, keep in mind that the wing loadings calculated by Eqs. (5.13–5.16), and Eqs. (5.22) and (5.32) are aerodynamic optimizations for only a portion of the mission. If these give wing loadings far lower than those in Table 5.5, they may be ignored.

When the best compromise for wing loading has been selected, the thrust-to-weight ratio should be rechecked to ensure that all requirements are still met. The equations in the last section which use T/W should be recalculated with the selected W/S and T/W . Only then can the next step of design, initial sizing, be initiated.

INITIAL SIZING

6.1 INTRODUCTION

Aircraft sizing is the process of determining the takeoff gross weight and fuel weight required for an aircraft concept to perform its design mission. Sizing was introduced in Chapter 3, in which a quick method based upon minimal information about the design was used to estimate the sizing parameters. That sizing method was limited to fairly simple design missions. This chapter presents a more refined method capable of dealing with most types of aircraft-sizing problems.

An aircraft can be sized using some existing engine or a new design engine. The existing engine is fixed in size and thrust, and is referred to as a “fixed-engine” (“fixed” refers to engine size).

The new design engine can be built in any size and thrust required, and is called a “rubber engine” because it can be “stretched” during the sizing process to provide any required amount of thrust.

Rubber-engine sizing is used during the early stages of an aircraft development program that is sufficiently important to warrant the development of an all-new engine. This is generally the case for a major military fighter or bomber program, and is sometimes the case for a transport-aircraft project such as the SST.

In these cases, the designer will use a rubber engine in the early stages of design, and then, with the customer, tell the engine company what characteristics the new engine should have. When the engine company finalizes the design for the new engine, it becomes fixed in size and thrust. The aircraft concept will then be finalized around this now-fixed engine.

Developing a new jet engine costs on the order of a billion dollars. Developing and certifying a new piston engine is also very expensive. Most aircraft projects do not rate development of a new engine, and so must rely on selecting the best of the existing engines. However, even projects which must use an existing engine may begin with a rubber-engine design study to determine what characteristics to look for in the selection of an existing engine.

The rubber engine can be scaled to any thrust so the thrust-to-weight ratio can be held to some desired value even as the aircraft weight is varied. The rubber-engine sizing approach allows the designer to size the aircraft to meet both performance and range goals, by solving for takeoff gross weight while holding the thrust-to-weight ratio required to meet the performance objectives. As the weight varies, the rubber-engine is scaled up or down as required.

This is not possible for fixed-engine aircraft sizing. When a fixed size engine is used, either the mission range or the performance of the aircraft must become a fallout parameter.

For example, if a certain rate of climb must be attained, then the thrust-to-weight ratio cannot be allowed to fall to an extremely low value. If the calculation of the takeoff gross weight required for the desired range indicates that the weight is much higher than expected, then either the range must be reduced or the rate of climb must be relaxed.

A typical example of this is the would-be homebuilder who got a good buy on a Lycoming O-320, and is designing an aircraft around that 150-hp engine. If the sizing results say that a bigger engine is required, the homebuilder will change the sizing requirements!

6.2 RUBBER-ENGINE SIZING

Review of Sizing

Chapter 3 presented a quick method of sizing an aircraft using a configuration sketch and the selected aspect ratio. From this information a crude estimate of the maximum L/D was obtained. Using approximations of the specific fuel consumption, the changes in weight due to the fuel burned during cruise and loiter mission segments were estimated, expressed as the mission-segment weight fraction (W_{i+1}/W_i). Using these fractions and the approximate fractions for takeoff, climb, and landing which were provided in Table 3.2, the total mission weight fraction (W_x/W_0) was estimated.

For different classes of aircraft, statistical equations for the aircraft empty-weight fraction were provided in Table 3.1. Then, the takeoff weight was calculated using Eq. (3.4), repeated below as Eq. (6.1).

Since the empty weight was calculated using a guess of the takeoff weight, it was necessary to iterate towards a solution. This was done by calculating the empty-weight fraction from an initial guess of the takeoff weight and using Eq. (6.1) to calculate the resulting takeoff weight. If the calculated takeoff weight did not equal the initial guess, a new guess was made somewhere between the two.

$$W_0 = \frac{W_{\text{crew}} + W_{\text{payload}}}{1 - (W_f/W_0) - (W_e/W_0)} \quad (6.1)$$

where

$$\frac{W_f}{W_0} = 1.06 \left(1 - \frac{W_x}{W_0} \right) \quad (6.2)$$

Equation (6.1) is limited in use to missions which do not have a sudden weight change, such as a payload drop. Also, in many cases Eq. (6.1) cannot be used for fixed-engine sizing.

Refined Sizing Equation

For missions with a payload drop or other sudden weight change, a slightly different sizing equation must be used. The takeoff weight is calcu-

Table 6.1 Empty weight fraction vs W_0 , A , T/W_0 , W_0/S , and M_{max}

$W_e/W_0 = (a + bW_0^{C1}A^{C2}(T/W_0)^{C3}(W_0/S)^{C4}M_{\text{max}}^{C5})K_{VS}$							
	a	b	$C1$	$C2$	$C3$	$C4$	$C5$
Jet trainer	0	4.28	-0.10	0.10	0.20	-0.24	0.11
Jet fighter	-0.02	2.16	-0.10	0.20	0.04	-0.10	0.08
Military cargo/bomber	0.07	1.71	-0.10	0.10	0.06	-0.10	0.05
Jet transport	0.32	0.66	-0.13	0.30	0.06	-0.05	0.05

K_{VS} = variable sweep constant = 1.04 if variable sweep
= 1.00 if fixed sweep

lated by summing the crew weight, payload weight, fuel weight, and empty weight. This is shown in Eq. (6.2), which resembles Eq. (3.1) except that the payload now includes a fixed payload and a dropped payload. The empty weight is again expressed as an empty-weight fraction, but the fuel weight is determined directly.

$$W_0 = W_{\text{crew}} + W_{\text{fixed payload}} + W_{\text{dropped payload}} + W_{\text{fuel}} + W_{\text{empty}} \quad (6.3)$$

or

$$W_0 = W_{\text{crew}} + W_{\text{fixed payload}} + W_{\text{dropped payload}} + W_{\text{fuel}} + \left(\frac{W_e}{W_0} \right) W_0 \quad (6.4)$$

As before, an initial guess of the takeoff weight is used to determine a calculated takeoff weight, and the solution is iterated until the two are approximately equal to within a few percent. Refined methods for determining the empty-weight fraction and fuel used are discussed below.

Empty-Weight Fraction

The empty-weight fraction is estimated using improved statistical equations. Tables 6.1 and 6.2 were prepared using data from Ref. 1 to provide

Table 6.2 Empty weight fraction vs W_0 , A , hp/W_0 , W_0/S , and V_{max} (mph)

$W_e/W_0 = a + bW_0^{C1}A^{C2}(\text{hp}/W_0)^{C3}(W_0/S)^{C4}V_{\text{max}}^{C5}$							
	a	b	$C1$	$C2$	$C3$	$C4$	$C5$
Sailplane—unpowered	0	0.75	-0.05	0.14	0	-0.30	0.06
Sailplane—powered	0	1.20	-0.04	0.14	0.19	-0.20	0.05
Homebuilt—metal/wood	0	0.69	-0.10	0.05	0.10	-0.05	0.17
Homebuilt—composite	0	0.59	-0.10	0.05	0.10	-0.05	0.17
General aviation—single engine	-0.25	1.14	-0.20	0.08	0.05	-0.05	0.27
General aviation—twin engine	-0.90	1.32	-0.10	0.08	0.05	-0.05	0.20
Agricultural aircraft	0	1.64	-0.14	0.07	0.10	-0.10	0.11
Twin turboprop	0.37	0.08	-0.06	0.08	0.08	-0.05	0.30
Flying boat	0	0.41	-0.01	0.10	0.05	-0.12	0.18

empty-weight equations which better reflect the weight impact of the major design variables. These are the aspect ratio, thrust-to-weight (or horsepower-to-weight) ratio, wing loading, and maximum speed.

The equations of Tables 6.1 and 6.2 result in a much better statistical fit, with only about half the standard deviation of the equations in Table 3.1. However, these equations should not be used to conduct design trade studies for one particular aircraft. That must be done using the component weight buildup methods in Chapter 15.

Fuel Weight

The remaining unknown in Eq. (6.4) is the fuel weight. Previously this was estimated as a fuel fraction by determining the ratio between the weight at the end of the mission and the takeoff weight (W_x/W_0). As the only weight loss during the mission was due to fuel usage, the fuel fraction was found simply as $(1 - W_x/W_0)$. This cannot be assumed if the mission includes a weight drop.

If the mission includes a weight drop, it is necessary to actually calculate the weight of the fuel burned during every mission leg, and sum for the total mission fuel. The mission segment weight fractions (W_i/W_{i-1}) are calculated as before for all mission segments other than those which are weight drops. For each mission segment, the fuel burned is then equal to:

$$W_{fi} = \left(1 - \frac{W_i}{W_{i-1}}\right) W_i \quad (6.5)$$

The total mission fuel, W_{fm} , then is equal to:

$$W_{fm} = \sum_1^x W_{fi} \quad (6.6)$$

The total aircraft fuel includes the mission fuel as well as an allowance for reserve and trapped fuel. This reserve fuel allowance is usually 5%, and accounts for an engine with poorer-than-normal fuel consumption. An additional allowance of 1% for trapped (i.e., unusable) fuel is typical. Thus, the total aircraft fuel is:

$$W_f = 1.06 \left(\sum_1^x W_{fi} \right) \quad (6.7)$$

The methods used for estimating the mission segment weight fractions are presented below. These are a combination of analytical and statistical methods, similar to the methods used in Chapter 3.

Engine Start, Taxi, and Takeoff

As before, the mission segment weight fraction for engine start, taxi, and takeoff is estimated historically. A reasonable estimate is:

$$W_1/W_0 = 0.97-0.99 \quad (6.8)$$

Climb and Accelerate

From data in Ref. 10, the weight fraction for an aircraft climbing and accelerating to cruise altitude and Mach number " M ," (starting at Mach 0.1), will be approximately as follows:

$$\text{Subsonic: } W_i/W_{i-1} = 1.0065 - 0.0325M \quad (6.9)$$

$$\text{Supersonic: } W_i/W_{i-1} = 0.991 - 0.007M - 0.01M^2 \quad (6.10)$$

For an acceleration beginning at other than Mach 0.1, the weight fraction calculated by Eqs. (6.9) or (6.10) for the given ending Mach number should be divided by the weight fraction calculated for the beginning Mach number using Eqs. (6.9) or (6.10).

For example, acceleration from Mach 0.1-0.8 requires a weight fraction of about 0.9805, whereas acceleration from Mach 0.1-2.0 requires a weight fraction of 0.937. To accelerate from Mach 0.8-2.0 would require a weight fraction of $(0.937/0.9805)$, or 0.956.

Cruise

Equation (3.6), repeated below as Eq. (6.11), is derived from the Breguet range equation for cruise as derived in Chapter 17. For propeller aircraft, the specific fuel consumption " C " is calculated from the propeller specific fuel consumption (C_p or C_{bhp}) using Eq. (3.10). Substitution of Eq. (3.10) into Eq. (6.11) yields Eq. (6.12).

$$\text{Jet: } \frac{W_i}{W_{i-1}} = \exp \frac{-RC}{V(L/D)} \quad (6.11)$$

where

- R = range
- C = specific fuel consumption
- V = velocity
- L/D = lift-to-drag ratio

$$\text{Prop: } \frac{W_i}{W_{i-1}} = \exp \left[\frac{-RC_{bhp}}{550 \eta_p (L/D)} \right] \quad (6.12)$$

where η_p = propeller efficiency.

During cruise and loiter, the lift equals the weight, so the L/D can be expressed as the inverse of the drag divided by the weight:

$$\frac{L}{D} = \frac{1}{\frac{qC_{D0}}{W/S} + \frac{W}{S} \frac{1}{q\pi Ae}} \quad (6.13)$$

Note that the wing loading used in Eq. (6.13) and subsequent weight fraction equations is the actual wing loading at the condition being evaluated, not the takeoff wing loading.

Loiter

Repeating Eq. (3.8), the weight fraction for a loiter mission segment is:

$$\text{Jet: } \frac{W_i}{W_{i-1}} = \exp \frac{-EC}{L/D} \quad (6.14)$$

where E = endurance or loiter time.

(Note—watch the units!) Substitution of Eq. (3.10) into Eq. (6.14) yields:

$$\text{Prop: } \frac{W_i}{W_{i-1}} = \exp \frac{-EVC_{bhp}}{550 \eta_p (L/D)} \quad (6.15)$$

Combat

The combat mission leg is normally specified as either a time duration (“ d ”) at maximum power (typically $d = 3$ min), or as a certain number of combat turns at maximum power at some altitude and Mach number. The weight of the fuel burned is equal to the product of thrust, specific fuel consumption, and duration of the combat, so the mission segment weight fraction is:

$$W_i/W_{i-1} = 1 - C(T/W)(d) \quad (6.16)$$

Note that the T/W is defined at combat weight and thrust, not at takeoff conditions. Again, watch the units, especially the time units.

If the combat is defined by some number of turns, the duration of combat (d) must be calculated. The time to complete “ x ” turns is the total number of radians to turn divided by the turn rate. When combined with Eq. (5.17), this yields:

$$d = \frac{2\pi x}{\dot{\psi}} = \frac{2\pi Vx}{g\sqrt{n^2 - 1}} \quad (6.17)$$

The load factor “ n ” for a sustained combat turn is found by assuming that the thrust angle is approximately aligned with the flight direction, so the thrust must equal the drag. The lift must equal the weight times the load factor n , which yields:

$$n = (T/W)(L/D) \quad (6.18)$$

This is subject to the constraints of maximum structural load factor [Eq. (6.19)] and maximum available lift [Eq. (6.20)].

$$n \leq n_{\max} \quad (6.19)$$

$$n \leq \frac{qC_{L\max}}{W/S} \quad (6.20)$$

The lift-to-drag ratio is found by including the load factor term into Eq. (6.13), which results in Eq. (6.21). The changes to C_{D0} and e at combat conditions which were discussed in the last chapter should be used in Eq. (6.21).

$$\frac{L}{D} = \frac{1}{q \frac{C_{D0}}{n(W/S)} + \frac{n(W/S)}{q\pi Ae}} \quad (6.21)$$

Descent for Landing

Descent is estimated historically:

$$W_i/W_{i-1} = 0.990 \text{ to } 0.995 \quad (6.22)$$

Landing and Taxi Back

Again, a historical approximation is used.

$$W_i/W_{i-1} = 0.992 \text{ to } 0.997 \quad (6.23)$$

Summary of Refined Sizing Method

The design and sizing method presented above, as summarized in Fig. 6.1, resembles in many respects the first-order method presented as Fig. 3.7,

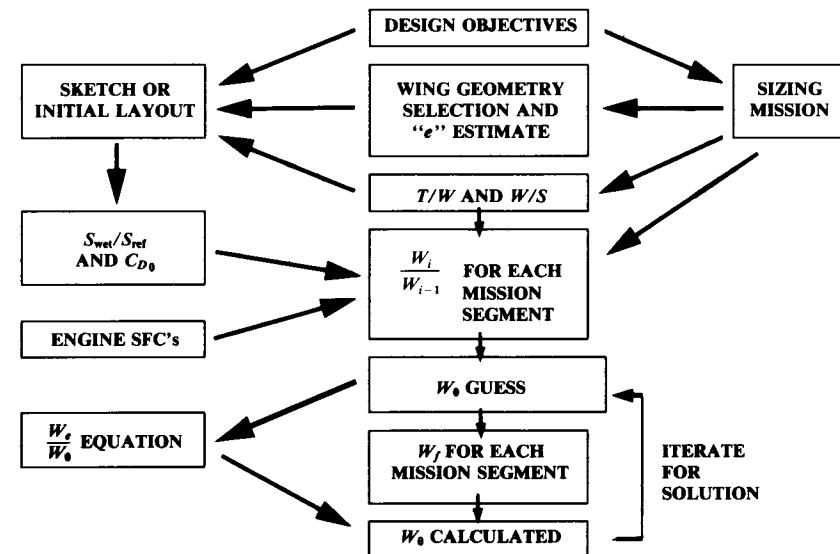


Fig. 6.1 Refined sizing method.

but makes use of more sophisticated analytical techniques and also permits sizing to missions which include weight drops.

From the design objectives and sizing mission, the wing geometry can be selected and an estimate of e obtained. A conceptual sketch or initial layout is used to estimate the wetted-area ratio, from which C_{D_0} is estimated. (Remember that e and C_{D_0} will be reduced during high-lift, combat conditions).

The methods of the previous chapter are used to select initial values for thrust-to-weight (or horsepower-to-weight) ratio and wing loading. Then the methods of this chapter are used along with engine data to determine the mission-segment weight fractions for each leg of the design mission.

The iteration for takeoff gross weight (" W_0 ") begins with an initial guess as to W_0 , and then the aircraft weight is calculated throughout the mission. For each mission leg, the aircraft weight will be reduced by either the weight of fuel burned or the payload weight dropped. Also, the total fuel burned is summed throughout the mission. Equations (6.7) and (6.4) are then used, along with a statistical empty weight fraction estimation, to arrive at a calculated W_0 .

If this does not equal the guessed value for W_0 , a new guess for W_0 is selected between the two values. Experience indicates that the solution will converge most rapidly if the new guess for W_0 is about three-fourths of the way from the initial guess to the calculated W_0 value.

This procedure is less complicated than it sounds. Examples can be found in Chapter 21.

(An alternative sizing method starts with a W_0 guess, and then subtracts the payload weight, crew weight, and calculated fuel weight to arrive at the "empty weight required" to perform the mission. This is compared to the statistical "empty weight available." If the empty weight required exceeds the empty weight available, then W_0 must be increased for the next iteration. This method is mathematically identical to the method presented above, but provides less-obvious guidance as to the next value of W_0 to use for iteration.)

6.3 FIXED-ENGINE SIZING

The sizing procedure for the fixed-size engine is similar to the rubber-engine sizing, with several exceptions. These result from the fact that either the mission range or the performance must be considered a fallout parameter, and allowed to vary as the aircraft is sized.

If the range is allowed to vary, the sizing problem is very simple. The required thrust-to-weight ratio (T/W) is determined as in the last section to provide all required performance capabilities, using the known characteristics of the selected engine. Then the takeoff gross weight is determined as the total engine takeoff thrust divided by the required takeoff thrust-to-weight ratio.

$$W_0 = \frac{NT_{\text{per engine}}}{(T/W)} \quad (6.24)$$

where N = number of engines.

With the takeoff weight known, the range capability can be determined from Eq. (6.4) using a modified iteration technique. The known takeoff weight is repeatedly used as the "guess" W_0 , and the range for one or more cruise legs is varied until the calculated W_0 equals the known W_0 .

This technique can also be used to vary mission parameters other than range. For example, a research aircraft may be sized for a certain radius (range out and back) with the number of minutes of test time as the variable parameter.

If some range requirement must be satisfied, then performance must be the fallout. The takeoff gross weight will be set by fuel requirements, and the fixed-size engine may not necessarily provide the thrust-to-weight ratio desired for performance considerations.

In this case the takeoff gross weight can be solved by iteration of Eq. (6.4) as for the rubber-engine case, with one major exception. The thrust-to-weight ratio is now permitted to vary during the sizing iterations. Equation (6.16) cannot be used for determining a weight fraction for combat mission legs as it assumes a known T/W .

Instead, the fuel burned during combat by a fixed-size engine is treated as a weight drop. For a given engine, the fuel burned during a combat leg of duration d is simply the thrust times the specific fuel consumption times the duration:

$$W_f = CTd \quad (6.25)$$

The weight of fuel calculated by Eq. (6.25) is treated as a weight drop in the iterations to solve Eq. (6.4). Once the takeoff gross weight is determined, the resulting thrust-to-weight ratio must be used to determine the actual aircraft performance for the requirements evaluated in the last chapter. If the requirements are not met, then either your aircraft design is not very good or the requirements are too tough!

6.4 GEOMETRY SIZING

Fuselage

Once the takeoff gross weight has been estimated, the fuselage, wing, and tails can be sized. Many methods exist to initially estimate the required fuselage size.

For certain types of aircraft, the fuselage size is determined strictly by "real-world constraints." For example, a large passenger aircraft devotes most of its length to the passenger compartment. Once the number of passengers is known and the number of seats across is selected, the fuselage length and diameter are essentially determined.

For initial guidance during fuselage layout and tail sizing, Table 6.3 provides statistical equations for fuselage length developed from data provided in Ref. 1. These are based solely upon takeoff gross weight, and give remarkably good correlations to most existing aircraft.

Fuselage fineness ratio is the ratio between the fuselage length and its maximum diameter. If the fuselage cross section is not a circle, an equivalent diameter is calculated from the cross-sectional area.

Table 6.3 Fuselage length vs W_0

Length = aW_0^C	a	C
Sailplane—unpowered	0.86	0.48
Sailplane—powered	0.71	0.48
Homebuilt—metal/wood	3.68	0.23
Homebuilt—composite	3.50	0.23
General aviation—single engine	4.37	0.23
General aviation—twin engine	0.86	0.42
Agricultural aircraft	4.04	0.23
Twin turboprop	0.37	0.51
Flying boat	1.05	0.40
Jet trainer	0.79	0.41
Jet fighter	0.93	0.39
Military cargo/bomber	0.23	0.50
Jet transport	0.67	0.43

Theoretically, for a fixed internal volume the subsonic drag is minimized by a fineness ratio of about 3.0 while supersonic drag is minimized by a fineness ratio of about 14. Most aircraft fall between these values.

A historically-derived fuselage fineness ratio can be used, along with the length estimate, to develop the initial fuselage layout. However, “real-world constraints” such as payload envelope must take priority. For most design efforts the realities of packaging the internal components will establish the fuselage length and diameter.

Wing

The actual wing size can now be determined simply as the takeoff gross weight divided by the takeoff wing loading. Remember that this is the reference area of the theoretical, trapezoidal wing, and includes the area extending into the aircraft centerline.

Tail Volume Coefficient

For the initial layout, a historical approach is used for the estimation of tail size. The effectiveness of a tail in generating a moment about the center of gravity is proportional to the force (i.e., lift) produced by the tail and to the tail moment arm.

The primary purpose of a tail is to counter the moments produced by the wing. Thus, it would be expected that the tail size would be in some way related to the wing size. In fact, there is a directly proportional relationship between the two, as can be determined by examining the moment equations presented in Chapter 16. Therefore, the tail area divided by the wing area should show some consistent relationship for different aircraft, if the effects of tail moment arm could be accounted for.

The force due to tail lift is proportional to the tail area. Thus, the tail effectiveness is proportional to the tail area times the tail moment arm. This

product has units of volume, which leads to the “tail volume coefficient” method for initial estimation of tail size.

Rendering this parameter nondimensional requires dividing by some quantity with units of length. For a vertical tail, the wing yawing moments which must be countered are most directly related to the wing span b_w . This leads to the “vertical tail volume coefficient,” as defined by Eq. (6.26). For a horizontal tail or canard, the pitching moments which must be countered are most directly related to the wing mean chord ($\bar{C}_w$). This leads to the “horizontal tail volume coefficient,” as shown by Eq. (6.27).

$$c_{VT} = \frac{L_{VT}S_{VT}}{b_w S_w} \quad (6.26)$$

$$c_{HT} = \frac{L_{HT}S_{HT}}{\bar{C}_w S_w} \quad (6.27)$$

Note that the moment arm (L) is commonly approximated as the distance from the tail quarter-chord (i.e., 25% of the mean chord length measured back from the leading edge of the mean chord) to the wing quarter-chord.

The definition of tail moment arm is shown in Fig. 6.2, along with the definitions of tail area. Observe that the horizontal tail area is commonly measured to the aircraft centerline, while a canard's area is commonly considered to include only the exposed area. If twin vertical tails are used, the vertical tail area is the sum of the two.

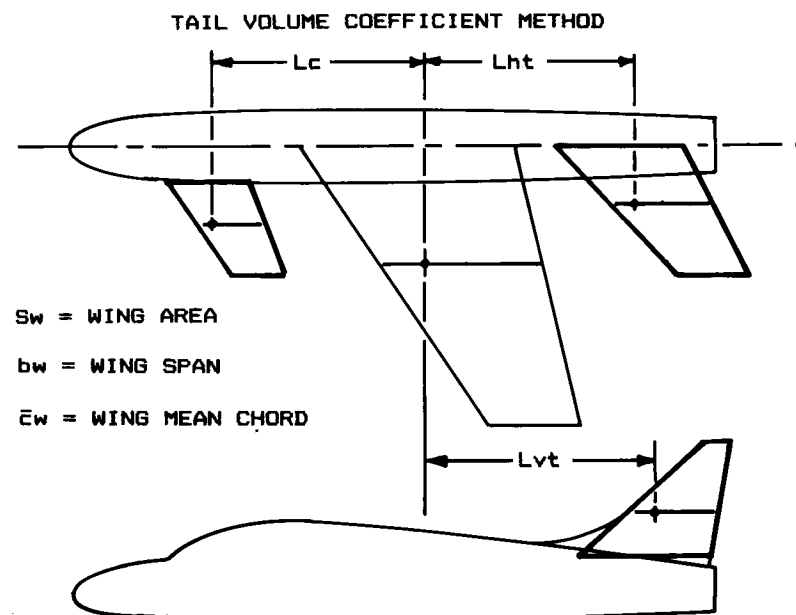


Fig. 6.2 Initial tail sizing.

Table 6.4 Tail volume coefficient

	Typical values	
	Horizontal c_{HT}	Vertical c_{VT}
Sailplane	0.50	0.02
Homebuilt	0.50	0.04
General aviation—single engine	0.70	0.04
General aviation—twin engine	0.80	0.07
Agricultural	0.50	0.04
Twin turboprop	0.90	0.08
Flying boat	0.70	0.06
Jet trainer	0.70	0.06
Jet fighter	0.40	0.07
Military cargo/bomber	1.00	0.08
Jet transport	1.00	0.09

Table 6.4 provides typical values for volume coefficients for different classes of aircraft. These values (conservative averages based upon data in Refs. 1 and 11), are used in Eqs. (6.28) or (6.29) to calculate tail area.

(Incidentally, Ref. 11 compiles a tremendous amount of aircraft data and is highly recommended for every designer's library.)

$$S_{VT} = c_{VT} b_W S_W / L_{VT} \quad (6.28)$$

$$S_{HT} = c_{HT} \bar{C}_W S_W / L_{HT} \quad (6.29)$$

To calculate tail size, the moment arm must be estimated. This can be approximated at this stage of design by a percent of the fuselage length as previously estimated.

For an aircraft with a front-mounted propeller engine, the tail arm is about 60% of the fuselage length. For an aircraft with the engines on the wings, the tail arm is about 50–55% of the fuselage length. For aft-mounted engines the tail arm is about 45–50% of the fuselage length. A sailplane has a tail moment arm of about 65% of the fuselage length.

For an all-moving tail, the volume coefficient can be reduced by about 10–15%. For a “T-tail,” the vertical-tail volume coefficient can be reduced by approximately 5% due to the end-plate effect, and the horizontal tail volume coefficient can be reduced by about 5% due to the clean air seen by the horizontal. Similarly, the horizontal tail volume coefficient for an “H-tail” can be reduced by about 5%.

For an aircraft which uses a “V-tail,” the required horizontal and vertical tail sizes should be estimated as above. Then the V surfaces should be sized to provide the same total surface area (Ref. 3) as required for conventional tails. The tail dihedral angle should be set to the arctangent of the square root of the ratio between the required vertical and horizontal tail areas. This should be near 45 deg.

The horizontal tail volume coefficient for an aircraft with a control-type canard is approximately 0.1, based upon the relatively few aircraft of this type that have flown. For canard aircraft there is a much wider variation in the tail moment arm. Typically, the canarded aircraft will have a moment arm of about 30–50% of the fuselage length.

For a lifting-canard aircraft, the volume coefficient method isn't applicable. Instead, an area split must be selected by the designer. The required total wing area is then allocated accordingly. Typically, the area split allocates about 25% to the canard and 75% to the wing, although there can be wide variation. A 50-50 split produces a tandem-wing aircraft.

For an airplane with a computerized “active” flight control system, the statistically estimated tail areas may be reduced by approximately 10% provided that trim, engine-out, and nosewheel liftoff requirements can be met. These are discussed in Chapter 16.

6.5 CONTROL-SURFACE SIZING

The primary control surfaces are the ailerons (roll), elevator (pitch), and rudder (yaw). Final sizing of these surfaces is based upon dynamic analysis of control effectiveness, including structural bending and control-system effects. For initial design, the following guidelines are offered.

The required aileron area can be estimated from Fig. 6.3, an updated version of a figure from Ref. 12. In span, the ailerons typically extend from

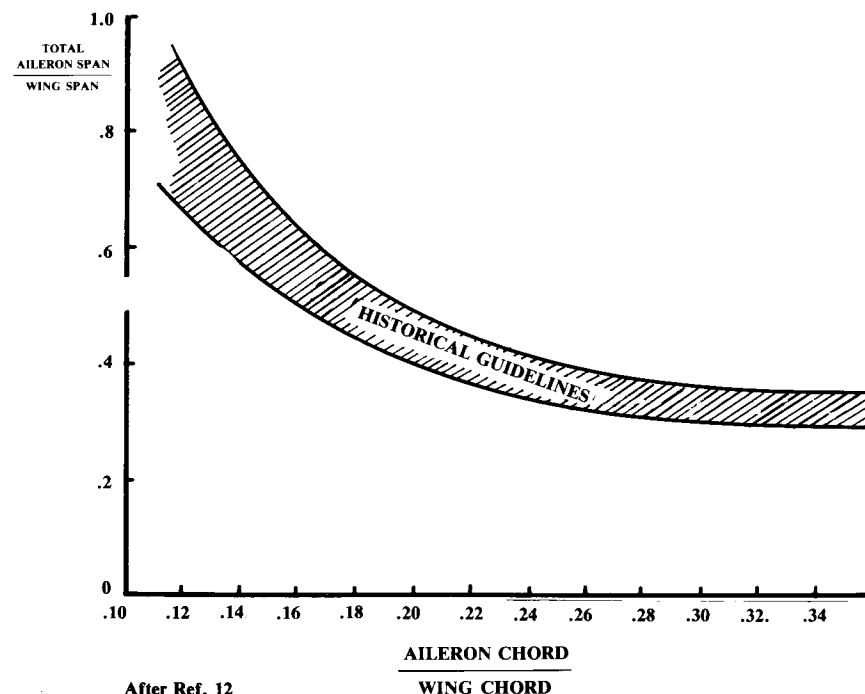


Fig. 6.3 Aileron guidelines.

about 50% to about 90% of the span. In some aircraft, the ailerons extend all the way out to the wing tips. This extra 10% provides little control effectiveness due to the vortex flow at the wingtips, but can provide a location for an aileron mass balance (see below).

Wing flaps occupy the part of the wing span inboard of the ailerons. If a large maximum lift coefficient is required, the flap span should be as large as possible. One way of accomplishing this is through the use of spoilers rather than ailerons. Spoilers are plates located forward of the flaps on the top of the wing, typically aft of the maximum thickness point. Spoilers are deflected upward into the slipstream to reduce the wing's lift. Deploying the spoiler on one wing will cause a large rolling moment.

Spoilers are commonly used on jet transports to augment roll control at low speed, and can also be used to reduce lift and add drag during the landing rollout. However, because spoilers have very nonlinear response characteristics they are difficult to implement for roll control when using a manual flight control system.

High-speed aircraft can experience a phenomena known as "aileron reversal" in which the air loads placed upon a deflected aileron are so great that the wing itself is twisted. At some speed, the wing may twist so much that the rolling moment produced by the twist will exceed the rolling moment produced by the aileron, causing the aircraft to roll the wrong way.

To avoid this, many transport jets use an auxiliary, inboard aileron for high-speed roll control. Spoilers can also be used for this purpose. Several military fighters rely upon "rolling tails" (horizontal tails capable of being deflected nonsymmetrically) to achieve the same result.

Elevators and rudders generally begin at the side of the fuselage and extend to the tip of the tail or to about 90% of the tail span. High-speed

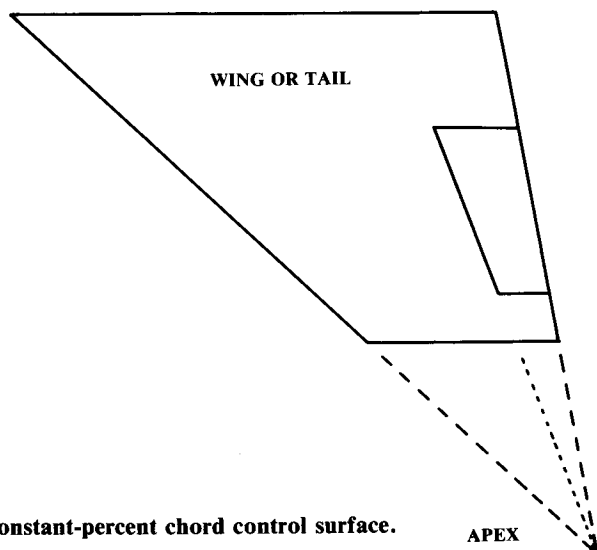


Fig. 6.4 Constant-percent chord control surface.

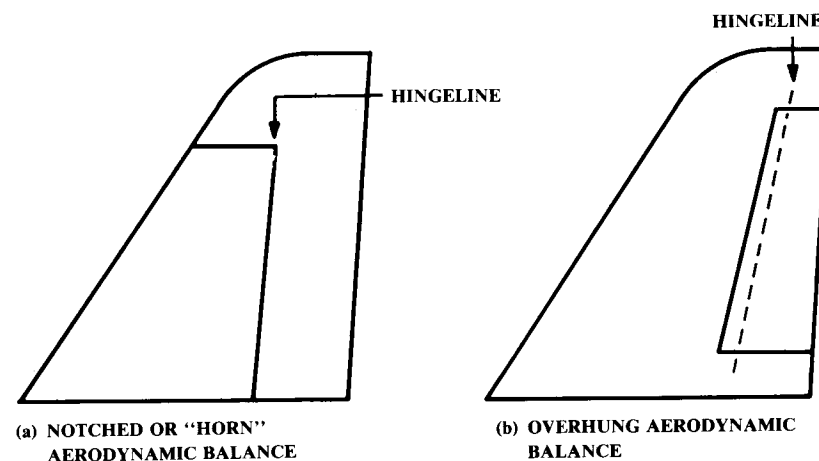


Fig. 6.5 Aerodynamic balance.

aircraft sometimes use rudders of large chord which only extend to about 50% of the span. This avoids a rudder effectiveness problem similar to aileron reversal.

Control surfaces are usually tapered in chord by the same ratio as the wing or tail surface so that the control surface maintains a constant percent chord (Fig. 6.4). This allows spars to be straight-tapered rather than curved. Ailerons and flaps are typically about 15–25% of the wing chord. Rudders and elevators are typically about 25–50% of the tail chord.

Control-surface "flutter," a rapid oscillation of the surface caused by the airloads, can tear off the control surface or even the whole wing. Flutter tendencies are minimized by using mass balancing and aerodynamic balancing.

Mass balancing refers to the addition of weight forward of the control-surface hingeline to counterbalance the weight of the control surface aft of the hingeline. This greatly reduces flutter tendencies. To minimize the weight penalty, the balance weight should be located as far forward as possible. Some aircraft mount the balance weight on a boom flush to the wing tip. Others bury the mass balance within the wing, mounted on a boom attached to the control surface.

An aerodynamic balance is a portion of the control surface in front of the hinge line. This lessens the control force required to deflect the surface, and helps to reduce flutter tendencies.

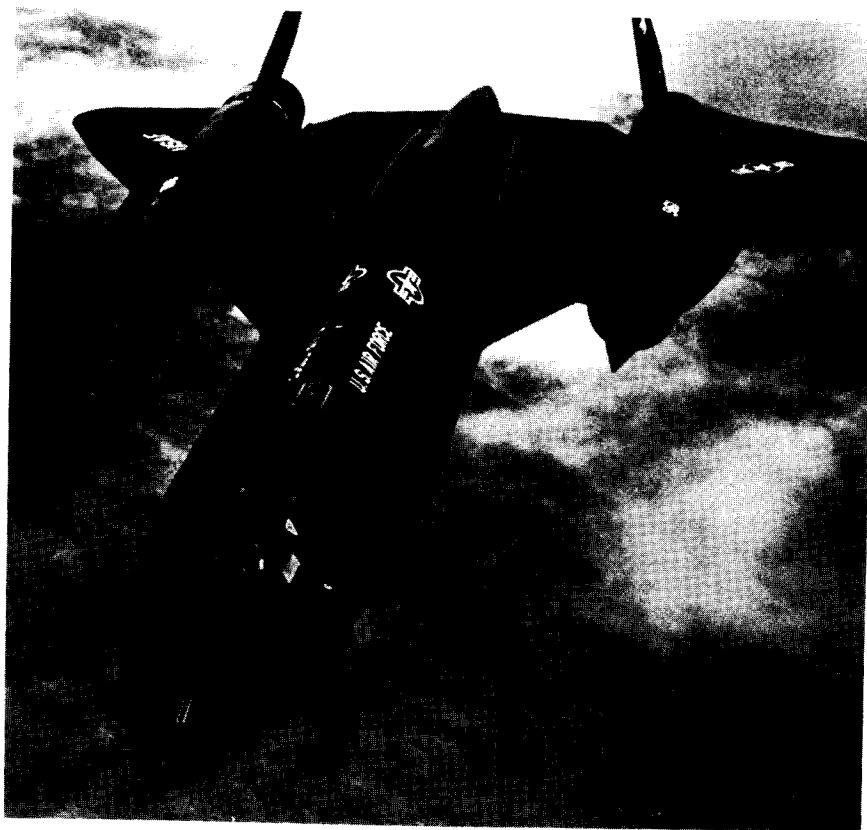
The aerodynamic balance can be a notched part of the control surface (Fig. 6.5a), an overhung portion of the control surface (Fig. 6.5b), or a combination of the two. The notched balance is not suitable for ailerons or for any surface in high-speed flight. The hinge axis should be no farther aft than about 20% of the average chord of the control surface.

The horizontal tail for a manually-controlled aircraft is almost always configured such that the elevator will have a hinge line perpendicular to the

aircraft centerline. This permits connecting the left- and right-hand elevator surfaces with a torque tube, which reduces elevator flutter tendencies.

Some aircraft have no separate elevator. Instead, the entire horizontal tail is mounted on a spindle to provide variable tail incidence. This provides outstanding "elevator" effectiveness but is somewhat heavy. Some general-aviation aircraft use such an all-moving tail, but it is most common for supersonic aircraft, where it can be used to trim the rearward shift in aerodynamic center that occurs at supersonic speeds.

A few aircraft such as the SR-71 have used all-moving vertical tails to increase control authority.



Lockheed SR-71

CONFIGURATION LAYOUT AND LOFT

7.1 INTRODUCTION

The process of aircraft conceptual design includes numerous statistical estimations, analytical predictions, and numerical optimizations. However, the product of aircraft design is a drawing. While the analytical tasks are vitally important, the designer must remember that these tasks serve only to influence the drawing, for it is the drawing alone that ultimately will be used to fabricate the aircraft.

All of the analysis efforts to date were performed to guide the designer in the layout of the initial drawing. Once that is completed, a detailed analysis can be conducted to resize the aircraft and determine its actual performance. This is discussed in Chapters 12-19.

This detailed analysis is time-consuming and costly, so it is essential that the initial drawing be credible. Otherwise, substantial effort will be wasted upon analyzing an unrealistic aircraft.

This chapter and Chapters 8-11 discuss the key concepts required to develop a credible initial drawing of a conceptual aircraft design. These concepts include the development of a smooth, producible, and aerodynamically acceptable external geometry, the installation of the internal features such as the crew station, payload, landing gear, and fuel system, and the integration of the propulsion system.

Real-world considerations which must be met by the design include the correct relationship between the aerodynamic center and the center of gravity, the proper amount of pilot outside visibility, and sufficient internal access for production and maintenance.

7.2 END PRODUCTS OF CONFIGURATION LAYOUT

The outputs of the configuration layout task will be design drawings of several types as well as the geometric information required for further analysis.

The design layout process generally begins with a number of conceptual sketches. Figure 7.1 illustrates an actual, unretouched sketch from a fighter conceptual design study (Ref. 13). As can be seen, these sketches are crude and quickly done, but depict the major ideas which the designer intends to incorporate into the actual design layout.

A good sketch will show the overall aerodynamic concept and indicate the locations of the major internal components. These should include the landing gear, crew station, payload or passenger compartment, propulsion sys-